

WOMEN SAFETY DEVICE

A Project Report

Submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of requirements for the award of degree

Bachelor of Technology

in

Electronics and Communication Engineering

by

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CERTIFICATE

This is to certify that the report entitled **WOMEN SAFETY DEVICE** submitted by **Sivapriya P** (NSS21EC084), **Sudharsana R** (NSS21EC089) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Electronics and Communication Engineering is a bonafide record of the project work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We hereby declare that the project report **WOMEN SAFETY DEVICE**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Dr. Lija Arun**

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

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Palakkad
14-06-2024

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Abstract

The Women Safety Device with GPS Tracking and Alert Messaging is an advanced system designed to enhance the safety of women by providing real-time monitoring and emergency response capabilities. It continuously monitors the user's heartbeat from activation, and if the heartbeat exceeds the normal range, indicating potential distress, the GSM module sends the user's current location, obtained from the integrated GPS module, to a pre-stored emergency contact. An emergency button allows for manual activation, sending a text message and placing an automatic call to the saved contact after a 10-second delay. During the call, the device activates its microphone to transmit surrounding sounds. This system automates the emergency response process, ensuring immediate action without requiring significant effort from the user, addressing the critical need for prompt assistance and enhancing personal safety in potentially dangerous situations.

Acknowledgement

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SIVAPRIYA P

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Chapter 1

Introduction

Ensuring the safety and security of women has become a critical concern in today's society. With increasing incidents of harassment and violence against women, there is an urgent need for innovative solutions that can provide immediate assistance and peace of mind. The development of an advanced Women Safety Device with GPS Tracking and Alert Messaging, specifically designed to address these safety concerns by leveraging modern technology. A tool created to assist women in need is a women's safety system. It is a wearable gadget that tracks the woman's whereabouts and, in the event of an emergency, sends a message to a saved emergency number along with emergency calling using electronic components such as GPS modules, microcontrollers, sensors, and wireless communication modules.

The gadget continuously monitors the heartbeat of the person wearing the device and initiates emergency calling and SMS. The device's GPS module uses satellite signals to pinpoint the woman's whereabouts, and it can also connect to the mobile network to exchange extra location-related data. In the present era, ensuring the safety of women has emerged as a critical concern due to the prevalence of physical and sexual abuse. This unfortunate reality creates an atmosphere of fear and restricts their ability to freely navigate public spaces.

A wearable security device designed for women's wrist contains an embedded microcontroller, emergency switch, GPS receiver, GSM modem, emergency switch and a mic. The device can alert parents and police of the user's current location via text message and gives call in case of an emergency. The device captures GPS location information, which is then used to generate a message containing location information and sent to the police and a pre-programmed mobile number. The GPS location can also be tracked on Google maps, allowing users to feel safe and secure.

When a woman feels unsafe, she can activate the system by using the start switch. This triggers the system to continuously monitor heart beat and woman's real-time location and a message to her registered contacts to inform them that she is in danger.

One of the most prevalent criminal offenses in the country currently is sexual harassment, which is increasing at an alarming rate. Women are known for their ability to mobilize people from different

backgrounds, such as ethnic, religious, political, and culture, for a common goal of promoting freedom. The project aims to contribute to the development of innovative solutions in addressing this significant issue.

The Women Safety Device project seeks to revolutionize personal security for women through the integration of cutting-edge technology. Central to its design are features such as GPS tracking, emergency alerts, and real-time monitoring, all aimed at providing rapid assistance in critical situations.

This compact device connects effortlessly to smartphones, allowing users to send immediate distress signals to pre-selected contacts and emergency services at the touch of a button. Its user interface is designed for simplicity and ease of use, ensuring accessibility for women of all ages and technological proficiency levels. Innovatively, the device can be embedded in wearable accessories like bracelets or keychains, offering a discreet yet powerful tool for personal safety. Leveraging artificial intelligence, the device can detect potential threats proactively, adding an additional layer of security. This makes it particularly beneficial for women navigating urban environments where the risk of harassment or assault is higher.

Beyond individual protection, the Women Safety Device has the potential to create a broader societal impact by fostering a sense of security and independence. It empowers women to move freely and confidently, knowing they have a reliable safety mechanism at their disposal. This project not only addresses immediate safety concerns but also contributes to long-term societal change by supporting the movement towards gender equality and safer public spaces for women.

Chapter 2

Literature Review

For the smooth completion of the miniproject, we have referred to various technical articles, research papers [1], [2], [3], [4] and blogs for a thorough understanding of the concerning issue and its related solutions.

The author proposes a security system designed to enhance women's safety and provide essential security measures. The system incorporates various colourful modules such as GPS guard (SIM900A), Atmega328 board, Arduino Board, GPS module (GYGPS6MV2), Screaming alarm (ADR 9600), Pressure detector, and power force unit. It helped to formulate the wearable technology [1].

An attempt is made to design a device named "Suraksha" to immediately alert the cops near the location from where the device is activated. which indeed provides worthy evidence against the crime. We formulated a device which sends location via a text message [2].

The authors have made an advancement in the present existing device which could send an emergency message to the contacts feed by the user. when the attacker or the culprit touches the watch a shock is induced which tends the attacker to let go of the victim. From this we devised a system that alerts the predefined contact [3].

The authors uniquely try to develop a wearable smart band with an additional secret webcam to record important information and it is connected to Bluetooth. Along with that the device also tracks the health of the person wearing it and monitors it using the device that's been connected. When the SOS button is pressed continuously having less or no delay, a message goes to the authorized person stating the person is in danger. We included a heart beat monitoring system to monitor changes in pulses of the victim [4].

Chapter 3

System Development

3.1 Hardware Components

- Power supply
- HT12E IC
- Arduino NANO
- HT12D IC
- 16x2 LCD Display
- RF Transceiver
- GPS NEO6MV2
- Push button
- GSM SIM900A
- 7805 IC (Voltage regulator)
- Heartbeat sensor
- Potentiometer
- Microphone

3.1.1 GPS Module

The NEO-6M GPS module is a popular GPS receiver with a built in ceramic antenna, which provides a strong satellite search capability. This receiver has the ability to sense locations and track up to 22 satellites and identifies locations anywhere in the world.

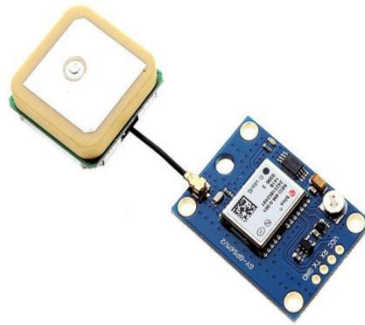


Fig 3.1.1 GPS Module

3.1.2 GSM Module

This is a GSM/GPRS-compatible Quad-band cell phone, which works on a frequency of 850/900/1800/1900MHz and which can be used for various applications such as access the Internet, make a voice call, send and receive SMS, etc. The frequency bands of the GSM modem can be set by AT Commands. The baud rate is configurable from 1200-115200 through AT command. The GSM/GPRS Modem is having an internal TCP/IP stack which enables us to connect with the internet via GPRS. This is an SMT type module and designed with a very powerful single-chip processor integrating AMR926EJ-S core, which is very popular in various industrial products.



GSM Module SIM900A

Fig 3.1.2 GSM Module

3.1.3 RF Transceiver

RF transceiver is a radio transmitter-receiver that can both send and receive digital signals using radio frequency (RF) waves. RF transceivers are made up of an antenna that receives signals and a tuner that separates specific signals from the others. The transceiver then demodulates the received signals and modulates them to receive new signals.

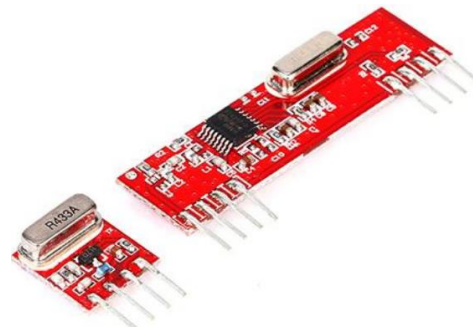


Fig 3.1.3 RF Transceiver

3.1.4 Arduino Nano

The Arduino Nano is an open-source breadboard-friendly microcontroller board based on the Microchip ATmega328P microcontroller (MCU) and developed by Arduino.cc and initially released in 2008. It offers the same connectivity and specs of the Arduino Uno board in a smaller form factor.

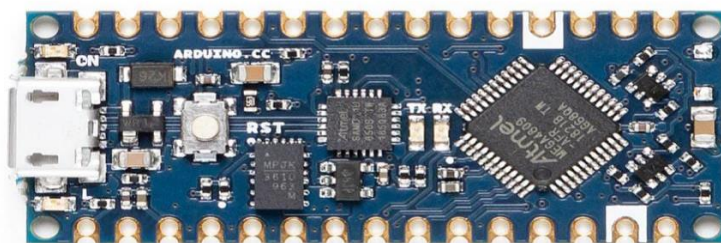


Fig 3.1.4 Arduino Nano

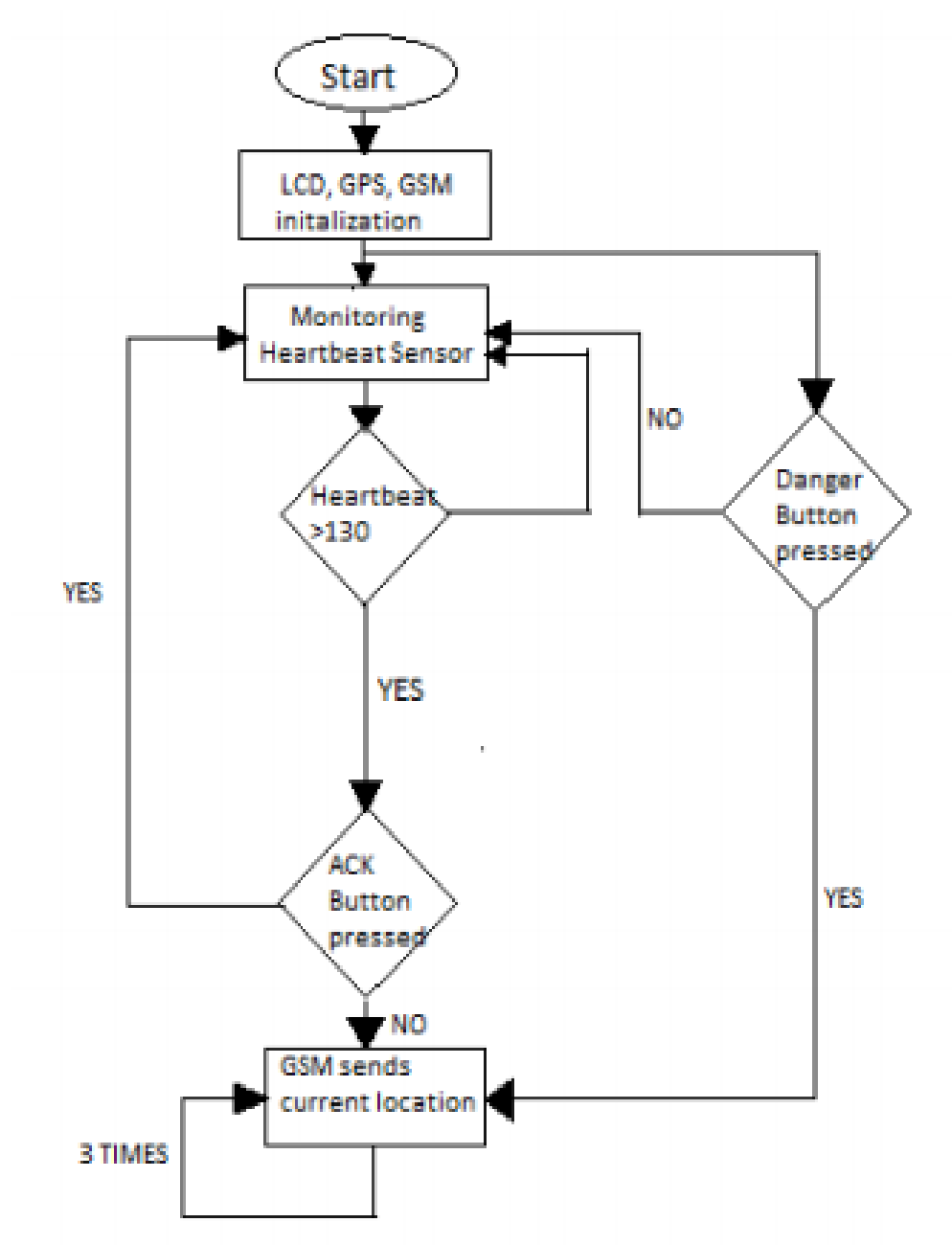
3.1.5 7805 Voltage Regulator

A 7805voltage regulator is an integrated circuit designed to maintain a constant output voltage despite changes in input voltage and load. Specifically, the 7805 is a positive voltage regulator that ensures a stable +5V output, making it a popular choice for powering various electronic components and circuits.



Fig. 3.1.5 7805 Voltage Regulator

3.2 Flow Chart



3.2 Flow Chart of Women Safety Device

Chapter 4

METHODOLOGY

Heart beat sensor data obtained is encoded by the encoder and transmitted via transmitter. On the other hand, the data is received at the receiver. The decoder decodes it and displays it on the LCD Screen. If the heartbeat is below 130, the heartbeat sensor keeps on monitoring the pulse-rate. If the heartbeat goes above the specified value i.e. normal heart rate, then the GSM module will send the help me message along with current location obtained from GPS module to the pre-stored contacts. Separately, a Danger button is also there, in case the person feels anything weird then she can directly press it. The message along with the location will be sent to the pre-stored contact. After 10s a call be sent to contact and once the call is attended nearby sounds can be heard. When the supply is switched on by connecting the wires, the heartbeat sensor starts monitoring the heartbeat. Location will be displayed on the LCD screen.

BLOCK DIAGRAM

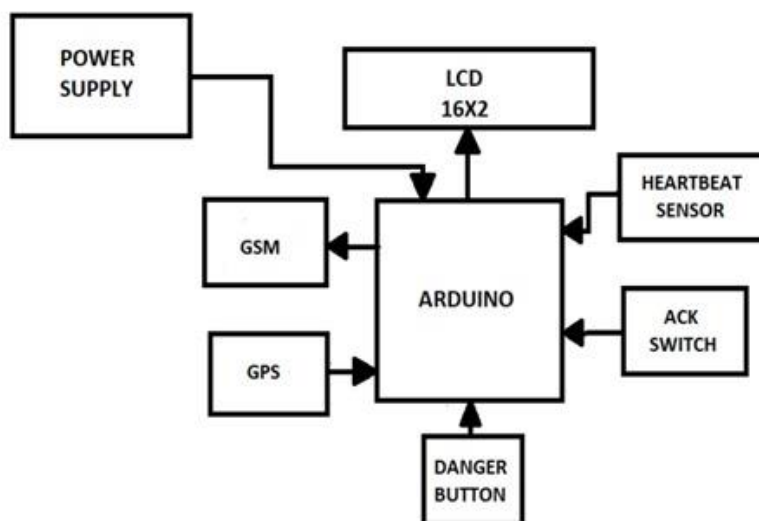


Fig 4.1 Block Diagram of Women Safety Device

Chapter 5

Results and Discussion

The mission of the project was successfully completed, and all project objectives were achieved. The primary outcomes are as follows:

- **Message Transmission:** Upon pressing the button, the system successfully tracked the current location of the victim and transmitted a message to the stored contact number via GSM. This message contained the location details of the victim, effectively notifying the recipient of the victim's location.
- **Safety Confirmation:** When the button was released, the system sent another message indicating that the person in danger is now safe. This functionality ensures that the recipient is promptly informed about the change in the victim's status.

These results demonstrate the effective implementation of the project's goals, providing a reliable means of communication during emergencies by utilizing GSM technology to send real-time updates about the victim's status and location.

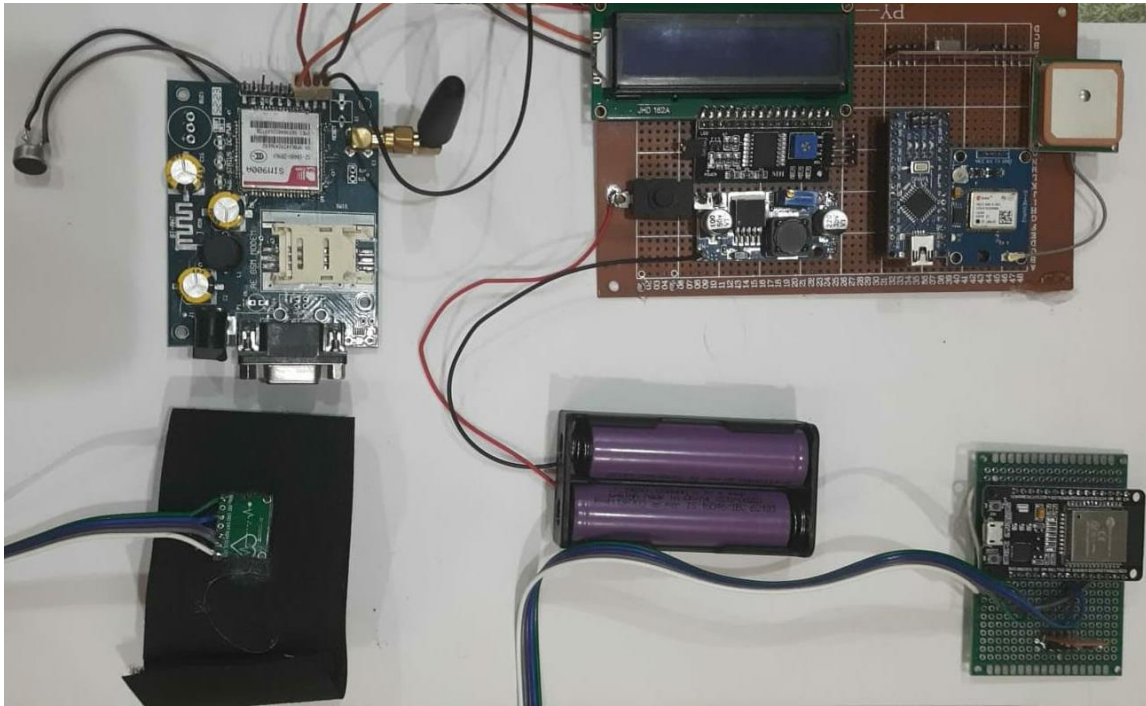


Fig 5.1 Hardware Circuit Model

SOURCE CODE

Source Code 1

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <TinyGPS++.h>
#include <SoftwareSerial.h>
#include <RCSwitch.h>
RCSwitch mySwitch = RCSwitch();
#define INTERVAL 5000
#define GPS_RX_PIN 11
#define GPS_TX_PIN 10
LiquidCrystal_I2C lcd(0x27, 16, 2);
TinyGPSPlus gps;
SoftwareSerial gpsSerial(GPS_RX_PIN, GPS_TX_PIN); // RX, TX

for GPS byte data=0; const int buttonPin = A0; // Use A0 as the
button pin bool buttonPressed = false; float latitude = 0.00000; float
longitude = 0.00000; // Phone number to send location const char*
phoneNumber = "8075607092";

void setup() { mySwitch.enableReceive(0); // Receiver on interrupt 0
=> that is pin #2

  lcd.init(); lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Women_safety");

  lcd.setCursor(0, 1);
  lcd.print("device_GPS_TRACK"); Serial.begin(9600);
  gpsSerial.begin(9600); // Initialize GPS serial communication
  pinMode(buttonPin, INPUT_PULLUP);

  delay(5000);
  lcd.clear(); lcd.setCursor(0, 0);
  lcd.print("location not_ found ");
  lcd.setCursor(0, 1);
}

void loop() {
  // Check if the button is pressed if
  (digitalRead(buttonPin) == LOW && !buttonPressed) {
```

```

buttonPressed = true;  sendLocation();  delay(100);
sms_only();  delay(100);  makeCall();  delay(100);
    }

    else if (digitalRead(buttonPin) == HIGH && buttonPressed) {
        // Reset button state when released
buttonPressed = false;

    }
    else if ((data==17)||((data==17))
    {
        sendLocation();
delay(100);
sms_only();
delay(100);
makeCall();
delay(100); data = false;

    }

    rf();
    lcd.setCursor(0, 1);
    lcd.print("status = ");
    lcd.print("Safe"); // Process GPS
    data  while (gpsSerial.available() >
0) {  if
(gps.encode(gpsSerial.read())) {
        // GPS data is available, you can access it through the gps object
    if (gps.location.isValid()) { // Valid GPS location data
        float latitude = gps.location.lat();    float longitude =
gps.location.lng();    // Display latitude and longitude on the
LCD
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Lat: ");
        lcd.print(latitude, 6);

```

```

lcd.setCursor(0, 1);
lcd.print("status = ");
lcd.print(data);
delay(600);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Long: ");
lcd.print(longitude, 6);
lcd.setCursor(0, 1);
lcd.print("status = ");
lcd.print(data);
delay(600);

    // Use latitude and longitude as needed
    Serial.print("Latitude: ");
    Serial.println(latitude, 6);
    Serial.print("Longitude: ");
    Serial.println(longitude, 6);
}
}
}
}
void rf()
{
    if (mySwitch.available())
    {
        // Serial.print("Received ");

data=(mySwitch.getReceivedValue() );

Serial.println(data);    delay(100);

mySwitch.resetAvailable();

    }
}
void sms_only()
{  lcd.clear();

lcd.setCursor(0, 0);

    Serial.println("Initializing GSM
connection...");    lcd.print("Initializing GSM
connection...");    delay(1000);    lcd.clear();

```

```

Serial.println("AT");    lcd.print("AT");

    delay(1000);
    lcd.clear();
    Serial.println("AT+CMGF=1"); // Set SMS mode to
text    lcd.print("AT+CMGF=1");    delay(1000);

    lcd.clear();
    lcd.print("AT+CMGS=\"");

    lcd.setCursor(0, 1);
    lcd.print(phoneNumber);
    Serial.print("AT+CMGS=\"");
    Serial.print(phoneNumber);

Serial.println("\n");

delay(1000);

    lcd.clear();


Serial.print("help_me");

delay(100);    lcd.clear();

    Serial.write(0x1A); // Send Ctrl+Z to end
message    Serial.println();    lcd.setCursor(0, 0);
    lcd.print("SMS_SEND");    lcd.setCursor(5, 1);
    lcd.print("help_me");
}
void makeCall() { // Initialize
GSM communication

    lcd.clear();

    lcd.setCursor(0, 0);

    lcd.print("Calling...");

    Serial.println("Initializing GSM connection...");
    delay(10000);

    Serial.println("AT");

    delay(1000);

    Serial.println("ATD" + String(phoneNumber) + ";"); // Dial the phone number
    delay(50000); // Adjust the duration of the call as needed

    Serial.println("ATH"); // Hang up the

```

```

call Serial.println("Call ended.");
lcd.clear(); lcd.setCursor(0, 0);
lcd.print("Call ended");
}
void sendLocation() { if
(gps.location.isValid()) {
    // Construct the message with location and map link
    String message = "Latitude: " + String(gps.location.lat(), 6) +
"\n";    message += "Longitude: " + String(gps.location.lng(), 6) +
"\n";

    message += "Map Link: https://www.google.com/maps?q=" + String(gps.location.lat(), 6) + ","
+ String(gps.location.lng(), 6);
    // Initialize GSM communication
    Serial.println("Initializing GSM
connection...");    delay(1000);
Serial.println("AT");    delay(1000);

    Serial.println("AT+CMGF=1"); // Set SMS mode to text
delay(1000);

    Serial.print("AT+CMGS=\"");
    Serial.print(phoneNumber);
Serial.println("\"");
delay(1000);

    Serial.print(message);
delay(100);

    Serial.write(0x1A); // Send Ctrl+Z to end message
    Serial.println();
lcd.clear();
lcd.setCursor(0, 0);
lcd.print(message);

    // Wait for response from GSM module    String response
= "";    unsigned long startTime = millis();    while (millis()
- startTime < 10000) { // Wait for 10 seconds
        if (Serial.available()) {
response += char(Serial.read());        if

```

```

(response.endsWith("OK\r\n")) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("SMS sent");
    Serial.println("SMS sent");
    return;
} else if (response.endsWith("ERROR\r\n")) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("SMS failed");
    Serial.println("SMS failed");
    return;
}
}
}

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("No response");
    Serial.println("No response");
} else {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Invalid location");
    Serial.println("Invalid location");
}
}

```

Source Code 2

```

/*
Example for different sending methods

```

```

https://github.com/sui77/rc-switch/

```

```

*/

```

```

#include <RCSwitch.h>

```

```

RCSwitch mySwitch = RCSwitch();
#define alrt 0 void

```



```

setup()
{
    Serial.begin(9600);

    // Transmitter is connected to Arduino Pin #10
    mySwitch.enableTransmit(2);
    pinMode(alrt,INPUT);

    // Optional set protocol (default is 1, will work for most outlets)
    // mySwitch.setProtocol(2);

    // Optional set pulse length.
    // mySwitch.setPulseLength(320);

    // Optional set number of transmission repetitions.
    // mySwitch.setRepeatTransmit(15);
}

void loop()
{
    if(digitalRead(alrt)==LOW)
    {
        /* See Example: TypeA_WithDIPSwitches
        */ mySwitch.switchOn("11111", "00010");

        delay(1000);

        mySwitch.switchOff("11111", "00010");

        delay(1000);
    }
}

```

Chapter 6

Conclusion

The proposed design seeks to tackle significant hurdles that women have faced in recent times and offer feasible remedies. These issues with the use of advanced equipment and innovative ideas. The solution for women's safety in India can be found in a cost-effective device that addresses women's challenges. This device enables sharing one's location with trusted contacts and creating a distress message to prevent further harm. This device can be made more compact and wearable by utilizing an Arduino Nano. Additionally, the device is designed to operate on low-powered rechargeable batteries, enhancing portability. In this study, the combination of GSM and Arduino technology has been employed to develop a women's safety system that aligns with the principles of smart cities. The system incorporates Arduino UNO, GSM, GPS, and sensors. In critical situations, the GSM and GPS components enable the victim's message and location to be transmitted to the authorized network, facilitating their quick and accurate identification. However, it is important to note that the system may face limitations in areas with limited connectivity. To overcome this challenge, future work should focus on integrating GPS location and transmitting danger messages to the surrounding public based on their respective locations.

With these tools, women can feel more secure when traveling to unfamiliar areas. While our current model is effective for tracking, there are other devices available that can serve as tracking tools. When activated with an emergency button, these apps can immediately notify a pre-selected contact list, including parents, friends, and the closest police station, about the victim's location. Similarly, voice detectors can also detect distress in the victim's voice and alert authorities accordingly. These technologies can be further improved in the future as well.

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